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# Robustness of Spoken Dialogue Processing Method Using Inductive Learning Based Amount of Information

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**ABSTRACT**
To be robustness, a spoken dialogue system must be able to process and accurately respond to conversation as well as adapting to background noise, interjections and unnecessary vocabulary. To do so the system needs to obtain information from the dialogue automatically. Therefore we propose a learning method using AoI(Amount of Information). We define the rule of spoken dialogue based on AoI. We can resolve background noise, interjections and unnecessary vocabulary using a response based on a large AoI. Our system produced a proper response rate of 65.3%. In this paper, we describe the experiment and discuss its results.

**KEY WORDS**
Spoken dialogue, Inductive Learning, Amount of Information

## 1 Introduction

Dialogue methods using a keyboard have been researched since the 1960's and many spoken dialogue systems now combine a speech recognition tool and a dialogue system using a keyboard. Examples using a keyboard include database systems—[1], systems with an unlimited domain[3], and systems using anaphora and semantics[4]. These systems either respond within a limited domain or continue a conversation with an unlimited domain. Systems operating within a limited domain are unable to respond to new situations and systems operating with an unlimited domain are rarely able to respond to the users command. In addition, typical spoken dialogues include background noise, interjections and unnecessary vocabulary.

Recently speech recognition techniques have improved with high rates of recognition. Spoken dialogue systems are now beginning to use web information. For example, JUPITER [2] provides weather information using a phone. JUPITER consists of speech recognition, language understanding and dialogue modeling. TOOT, which is another spoken dialogue system, can respond with information concerning train timetables. These systems use a frequently updated data of limited domain. However if the systems are unable to obtain the necessary information from the web, they will not be able to respond appropriately. As it is hard to find individual personal information on the web, the system needs to obtain information from a dialogue automatically. Therefore the system needs to have a learning process. Kimura et al. have proposed a spoken dialogue method of Inductive Learning using Genetic Algorithm(GA-IL)[7]. The most important aspect of GA-IL is its acquisition capability. GA-IL method can acquire the rules of spoken dialogue from a dialogue example. The main capability of learning is to distinguish common parts and different parts. The common parts mean the matching parts when the system compares the rules of spoken dialogue[refer 2.1]. However GA-IL cannot efficiently acquire the rules if the spoken dialogue contains few matching rules. Therefore we propose a learning method using "Amount of Information" (hereafter abbreviated as AoI).

We define the common parts and the different parts based on AoI. We try to deal with background noise, interjections and unnecessary vocabulary. We can resolve these problems using a response based on a large AoI. In this paper, we confirm that our system can robustly acquire response rules and is able to make a reply.

## 2 Common and different parts

### 2.1 Common and different parts based on GA-IL

We describe the definition of the common and different parts based on GA-IL. Here, we refer to the dialogue example of Table 1 which is constructed from user A and user B utterance. Greek characters are used in the example because human beings cannot understand

Table 1. Example of dialogue.

| User | Number | Utterance |
|---|---|---|
| A | 1 | $\alpha\delta\nu\beta\epsilon$ |
| B | 2 | $\kappa\varepsilon\zeta\vartheta\eta$ |
| A | 3 | $\alpha\delta\Gamma\pi\epsilon$ |
| B | 4 | $\xi\sigma\zeta\vartheta\eta$ |

Table 2. How to acquire common parts and different parts using GA-IL.

| Spoken dialogue rules | |
|---|---|
| Left hand side | Right hand side |
| $\underline{\alpha\delta}\ \nu\beta\ \underline{\epsilon}$ | $\kappa\varepsilon\ \underline{\zeta\vartheta\eta}$ |
| $\underline{\alpha\delta}\ \Gamma\pi\ \underline{\epsilon}$ | $\xi\sigma\ \underline{\zeta\vartheta\eta}$ |
| Pairs of the different parts | |
| $\nu\beta$ | $\kappa\varepsilon$ |
| $\Gamma\pi$ | $\xi\sigma$ |
| Common parts + various | |
| $\alpha\delta$ *@diff* $\epsilon$ | *@diff* $\zeta\vartheta\eta$ |

*@diff* is various.

the meaning of words when they do not know the language. The relationship between user A utterance and user B utterance is important. Therefore the rules acquired from the dialogue are based upon this relationship. The GA-IL system seeks the common and different parts from the rules. In Table 2, the common parts are the matching parts, which are underlined, and the others parts are the different parts. The pairs of the different parts have a characteristic relationship. theses pairs are **"response rules"**. The rules, which replace the different part with a variable, are **"generation rules"** and act as templates for making a sentence. We decided a condition of acquisition is that

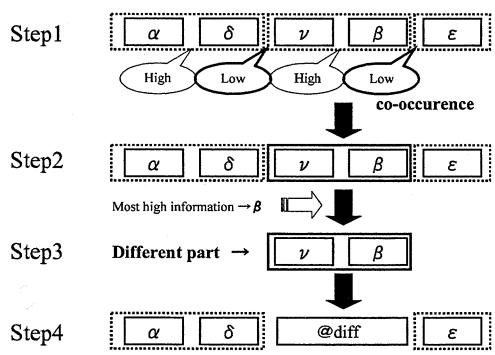



Figure 1. Three pattern of division.

the GA-IL system finds only one different part pair in each rule. The GA-IL system acquires the relationship, being the different part pair, which is obviously important. The pair of the different part is characterized by their relationship.

## 2.2 Common and different parts based on AoI

We define the common and different parts based on AoI. The common part consists of a small AoI because the words of common parts frequently exist in the dialogue example. The different part includes a large AoI because of characteristic parts.

We describe how to decide these parts using Figure 1. The division points of the sentence are decided by co-occurrence. We define that the number of division parts is three because we have used this number in the previous method. Therefore the division points are two of the lowest co-occurrence in the sentence. In the case of Figure 1, the division points are the first point between $\delta$ and $\nu$ , the second point between $\beta$ and $\varepsilon$. If $\beta$ is the largest AoI, the different part is $\nu\beta$.

# 3 Process

## 3.1 Learning process

The system acquires the response rule and the generation rule. The response rule is to select the response content, and the generation rule makes a sentence using a selected rule. Below we describe how these rules are acquired.

### 3.1.1 Response rule

The response rule is the pair of different parts between user A utterance and user B utterance. The different part is selected from the largest AoI of the three parts in the sentence. The AoI is as follows

AoI $I(x) = -logP(x) \cdots (1)$

$P(x) = \frac{F(x)}{N}$, $F(x)$ is the frequency of word x, N is sum of word number. According to the formula (1), low frequency indicates a large AoI. A response rule consists of both a left hand side(LHS), which includes the previous different part and the current different part, and a right hand side(RHS) which is the different part for the response. The system makes a sentence by combining the RHS of the response rule and the generation rule.

### 3.1.2 Generation rule

A generation rule consists of common parts and a variable which replaces the different parts. The generation

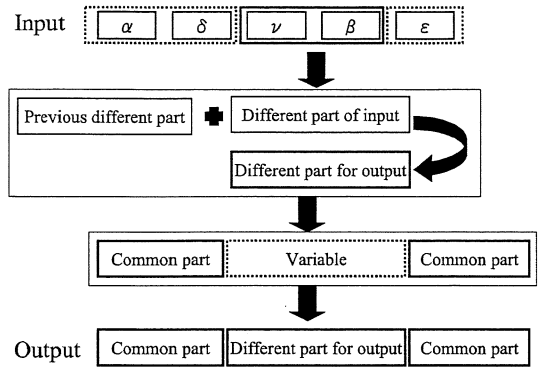



Figure 2. Flow of response generation.

rule then exchanges the variable for the RHS of the selected response rule.

### 3.1.3 How to use rules

Figure 2 shows how to use the rules. We assume that an input sentence is " $\alpha\delta\nu\beta\epsilon$ ". The system divides the input sentence into three parts which consist of " $\alpha\delta$ / $\nu\beta$ / $\epsilon$ " using co-occurrence, and the system decides the different part. The different part of the input sentence is $\nu\beta$. The system compares the different part of the input sentence with the LHS of response rules. If the system finds the matching rule, the system combines the RHS of response rule with the generation rule. If the system finds many matching rules, the system decides using information of the previous different part.

## 3.2 Dialogue processing

Figure 3 shows the outline of dialogue processing. The system recognizes the user's utterance using speech recognition tool[11]. The system extracts the different parts from the recognition, and selects a response rule. The selected response rule is combined with the generation rule. The combined generation rule is the response to the input. If the system could not make a response, the system uses either the relationship of the common parts or the smallest AoI. The User is given the response by a speech synthesis tool [12]. We explain each process.

### 3.2.1 Extraction of different part

A input sentence is divided into words by JUMAN[10]. The system calculates word frequency and co-occurrence. The division point is decided by co-occurrence. The different part is the largest AoI in the divided parts.

### 3.2.2 Selection of response rule

The response rule is selected by comparing the different parts of the user utterance with each rule in the response rule dictionary. If there are many matching rules, the system selects the rule using the previous different parts.

### 3.2.3 Selection of generation rule

A generation rule is just like a template, which can substitute the variable. When the generation rule is combined with the different part which it replaced the variable, the system uses mutual information. Mutual information is shown in formula (2).

$$M(x,y) = log\frac{P(x,y)}{P(x)P(y)} \cdots (2)$$

$P(x,y) = \frac{F(x,y)}{N}, P(x) = \frac{F(x)}{N}, P(y) = \frac{F(y)}{N}$. $F(x,y)$ is the co-occurrence between x and y, $F(x)$ is frequency of x, $F(y)$ is frequency of y and N is the total frequency of all words. Our new method combines Inductive Learning with statistics.

### 3.2.4 Response of small AoI

The response of small AoI is used for continuing a conversation. In human conversation, even if a speaker could not understand his/her partner's utterance, the speaker would reply giving a small AoI. Using a small AoI to respond to a large AoI means the details of the topic or information are unknown. A large AoI is described in 3.2.1, 3.2.2 and 3.2.3 as the different parts. If the system could not find the appropriate different part to respond to a large AoI, the system replies using a small AoI.

### 3.2.5 Response using common part

Though the response of a small AoI provides a vague answer, the system tries to keep the consistency of conversation using common parts. We consider that the response using common parts is better than the response using a small AoI.

# 4 Evaluation experiment of acquired rules

## 4.1 Purpose of experiment

The experiment purpose is to evaluate the acquired rules obtained from the common and different parts of ATR's Speech and Language Database(SLDB). SLDB is made by using two languages which consist of both English and Japanese, and it has same content dialogue examples of each language. Our system has an

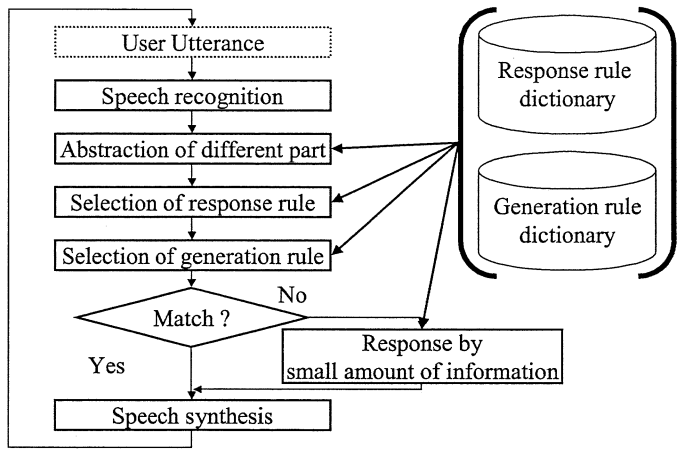



Figure 3. Flow of dialogue process.

Table 3. Example of the acquired different rules.

| | | Data example. |
|---|---|---|
| Clerk | 1 | Very good. May I have the spelling of your name, please? |
| Customer | 2 | uh Yes. It's T- A- N- A- K- A. |
| | | Example of word frequency and co-occurrence. |
| Clerk | 1 | (Very:29){6}(good.:10)**{1}**(May:40){39}(I:435){86}(have:261){21}(the:811)**{2}** (spelling:4){3}(of:185){8}(your:225){12}(name,:14){10}(please?:77) |
| Customer | 2 | (uh:127){6}(Yes.:64){3} (It's:31)**{1}**(T-:2){2}(A-:12){2}(N-:3){2}(A-:12){2}(K-:5){2}(A.:2) |
| | | Example of the acquired different parts. |
| Clerk | 1 | spelling of your name, please? |
| Customer | 2 | T-A-N-A-K-A. |

independent of learning capability of language. In order to verify the independent learning capability, we evaluate acquired rules each language.

## 4.2 Experiment procedure

The experiment data includes 1,576 sentences that consists of the conversation between a htel clerk and a customer.The system acquires the common and different parts from the SLDB of each language. The acquired different parts are evaluated by two people, a man and a woman. The evaluated data consists of 200 acquired rules.

## 4.3 Results and consideration

Table 3 shows an example of acquired different parts. We explain the acquisition process by this example. In (Very:29){6}(good.:10) of the Table 3, "29" shows frequency of "Very", "10" shows frequency of "good." and {6} shows co-occurrence between "Very" and "good.". Both "(good.:10)**{1}**(May:40)" and "(the:811)**{2}**(spelling:4)" are the split points in the sentence because co-occurrence is the lowest point in all division points. Table 4 and Table 5 show the relationship of the acquired different parts each language. We consider that it is possible to understand the content of dialogue using the different parts because different parts include important phrases. In the case of the spoken dialogue example, the system needs to judge important parts because the example includes noise, interjections and unnecessary vocabulary. For example, when the hotel clerk asks "Very good. May I have the spelling of your name, please?", the important part is "the spelling of your name, please?". The customer's answer should include the spelling of his name "TANAKA",T-A-N-A-K-A. Furthermore as shown Table 4 and Table 5, the system could acquire the similar meaning pair between the clerk and the customer in two languages. Our system could obtain suitable rules without depending on a language. Table 6 shows the acquired generation rules. These acquired

generation rules can be used in various situations.

Table 7 shows the evaluation of acquired important parts in Japanese. The two Japanese subjects evaluated the important parts selected by our system and determined that almost 60% were correct[1]. We consider that this rate is a very high because this experiment data includes background noise, interjections and unnecessary vocabulary.

Table 4. Example of the relationship of acquired different rules in English.

| | |
|---|---|
| Clerk | spelling of your name, please? |
| Customer | T-A-N-A-K-A. |
| Clerk | the type of card? |
| Customer | card is a VISA card. |
| Clerk | May I have your name, |
| Customer | Harris. |

Table 5. Example of the relationship of acquired different rules in Japanese.

| | |
|---|---|
| Clerk | スペルを |
| Customer | ティーエーエヌエーケイエー |
| Clerk | カードの種類 |
| Customer | ビザカードですけど。 |
| Clerk | お客様のお名前 |
| Customer | エイミー・ハリス |

Table 6. Example of the acquired template rules.

| | | |
|---|---|---|
| uh Yes. It's | @diff | |
| oh, | @diff | floor. |
| Okay, | @diff | sir? |
| | @diff | please? |
| Yes, that's | @diff | |
| Hello, | @diff | May I help you? |
| Very good. And, | @diff | |

## 5 Spoken dialogue experiment

### 5.1 Experiment procedure

The purpose is to evaluate the system with real spoken dialogue(open data).

[1]We evaluated only Japanese rules.

The learning data is based upon conversations with free topics between two people. The participants are three university students. The system acquired the rules by listening to the conversation between a pair of speakers. We call the three speakers, C,D and E respectively. After the system has acquired the rules, each speaker individually communicates with the system. Each response is divided into 4 patterns, being correct, semi correct, vague and erroneous. The correct response is a satisfactory response. The semi correct response is comprehensible, but not a satisfactory response. The vague response has some meanings, but able to understand. The erroneous response is not understood.

Table 7. Evaluation of acquired rules in Japanese.

| Evaluation of different part. | | |
|---|---|---|
| Subjects | A | B |
| Matching number | 111 | 121 |
| Matching rates | 55.5% | 60.5% |

Table 8. Result of response patterns.

| Speaker | C | D | E |
|---|---|---|---|
| Correct | 6.12% | 17.02% | 17.89% |
| Semi Correct | 31.36% | 22.34% | 32,36% |
| Vague | 27.55% | 11.70% | 3.16% |
| Erroneous | 34.69% | 48.94% | 46.32% |

### 5.2 Experiment result and consideration

Table 8 shows the results being the response patterns as determined by each speaker. The proper response rates show a comprehension between 51.0% and 65.3%(by proper responses we mean Correct Response, Semi Correct Response and Vague Response). In open data of spoken dialogue, these results show a high success response rates because the spoken dialogue includes erroneous recognitions and the user's erroneous utterance.

## 6 Conclusion

Our purpose is to create a spoken dialogue system which is able to cope with frequent changes in conversation topics as occurs in daily life. Therefore we proposed a learning method using AoI(Amount of Information). In this paper, we extended the definition

of common and different parts by AoI, co-occurrence and mutual information.

In evaluating the experiment and the acquired rules, the two Japanese subjects evaluated the important parts selected by our system and determined that almost 60% were correct. In the spoken dialogue experiment, our system produced a proper response rate of 65.3%. These results suggested that our system was able to deal with spoken dialogue, which included noise, various interjections and unnecessary vocabulary.

Future work, we will improve division points and correct response rate.

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